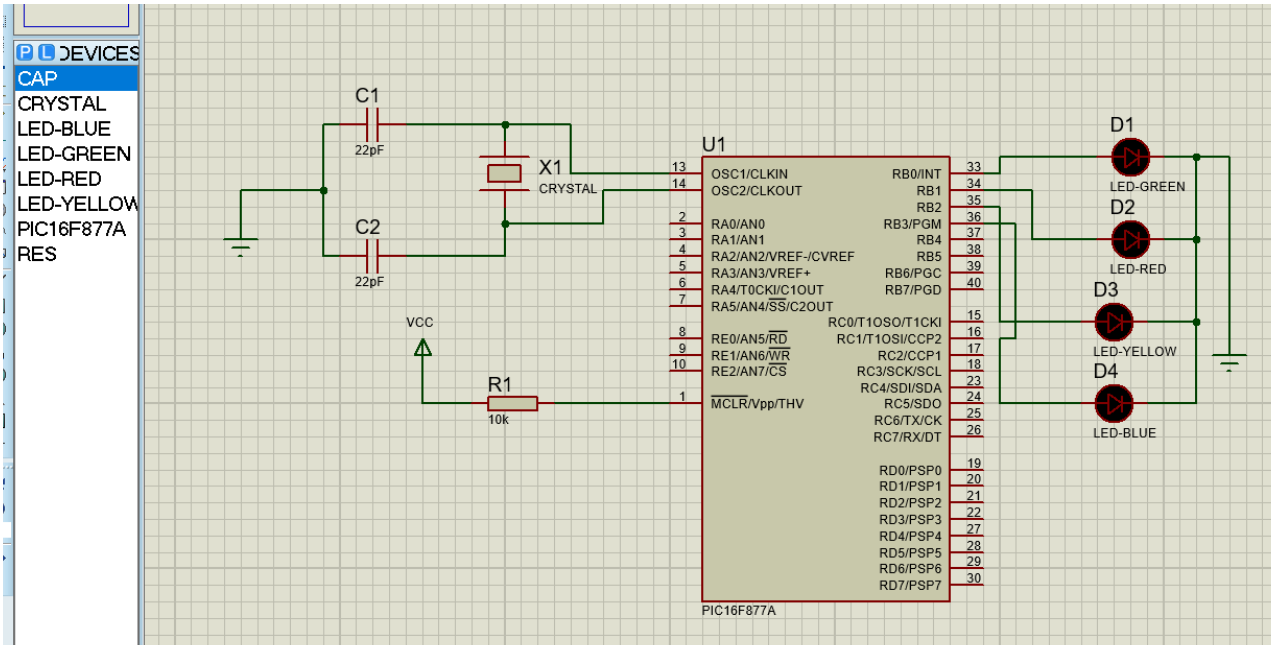
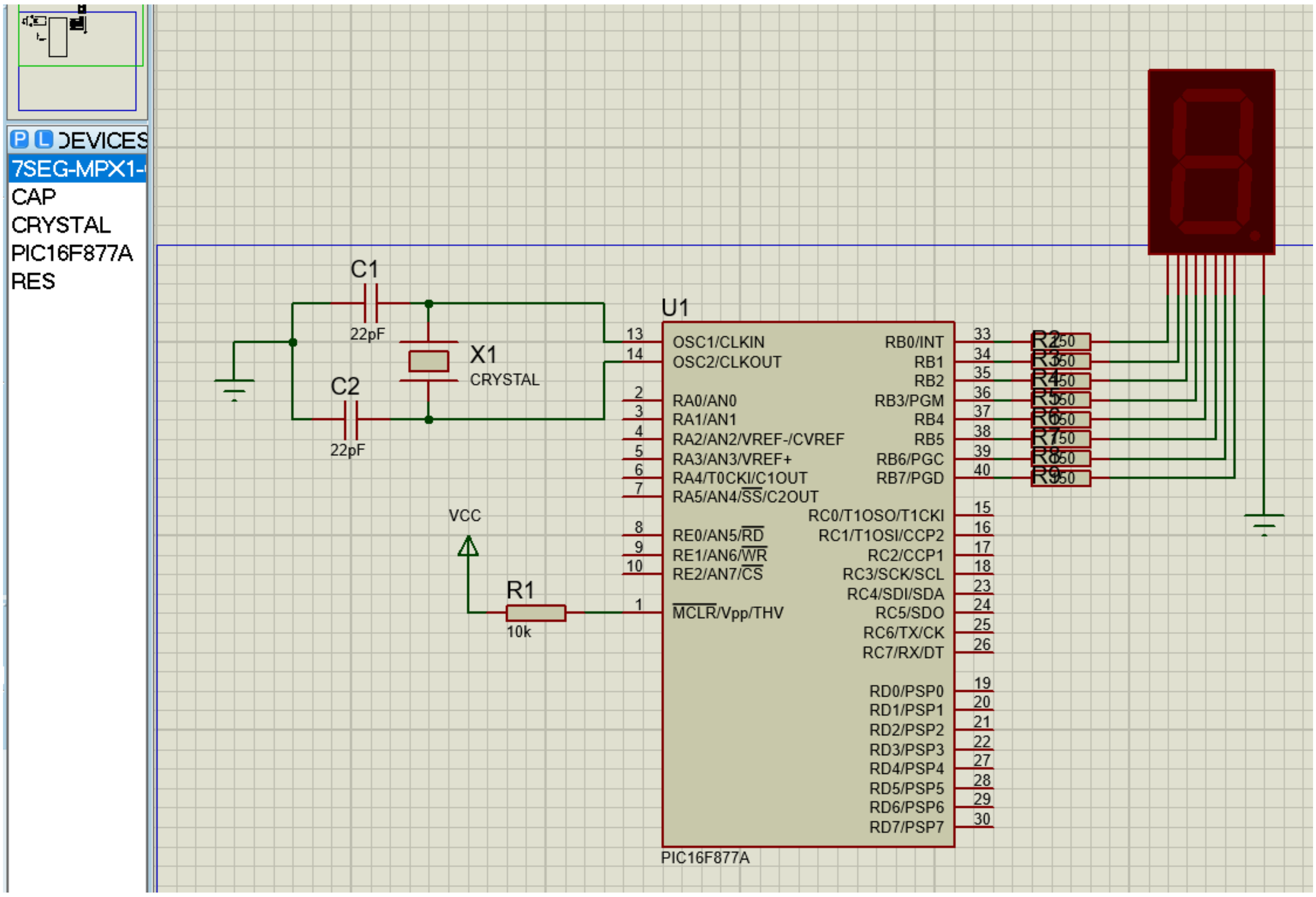
Circuit diagram:



Source Code :

```
void main() {
  TRISB = 0x00;   // Configure pin RB0 (pin 33) as an output pin
  PORTB = 0x00;  // Ensure the LED is initially off

  while(1)       // Infinite loop
  {
    PORTB.F0 = 1;
    PORTB.F1 = 0;// Turn LED ON (set pin RB0 high)
    PORTB.F2 = 0;
    PORTB.F3 = 0;
    Delay_ms(300); // Delay for 1000 milliseconds (1 second)
    PORTB.F0 = 0;
    PORTB.f1 = 1;  // Turn LED OFF (set pin RB0 low)
    PORTB.F2 = 0;
    PORTB.F3 = 0;
    Delay_ms(300); // Delay for 1000 milliseconds (1 second)
     PORTB.F0 = 0;
    PORTB.F1 = 0;// Turn LED ON (set pin RB0 high)
    PORTB.F2 = 1;
    PORTB.F3 = 0;
      Delay_ms(300);
     PORTB.F0 = 0;
    PORTB.F1 = 0;// Turn LED ON (set pin RB0 high)
    PORTB.F2 = 0;
    PORTB.F3 = 1;
    Delay_ms(300); }}
```

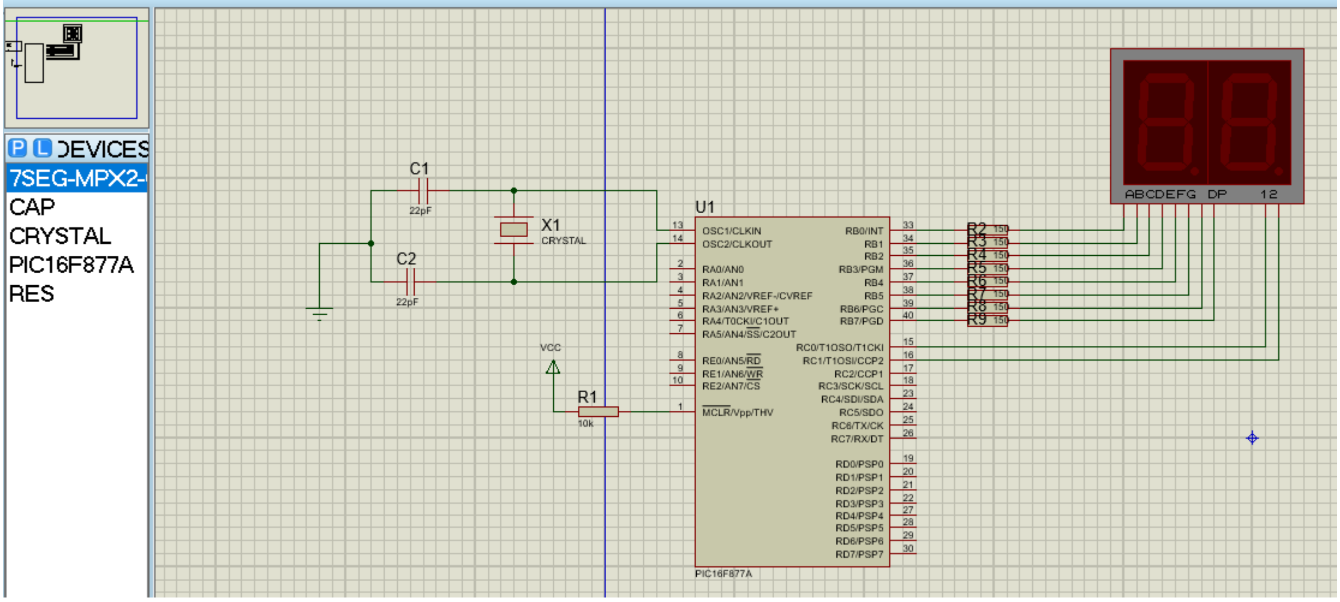
Circuit Diagram:



MikroC Code :

```
char array[]={0x3f, 0x06, 0x5b, 0x4f, 0x66, 0x6d, 0x7d, 0x07, 0x7f, 0x6f};
 int i;
void main() {

      TRISB=0x00;
      portb=0x00;
      while(1)
      {
         for(i=0;i<10;i++)
         {
         portb=array[i];
          delay_ms(1000);
       }
      }

}
```

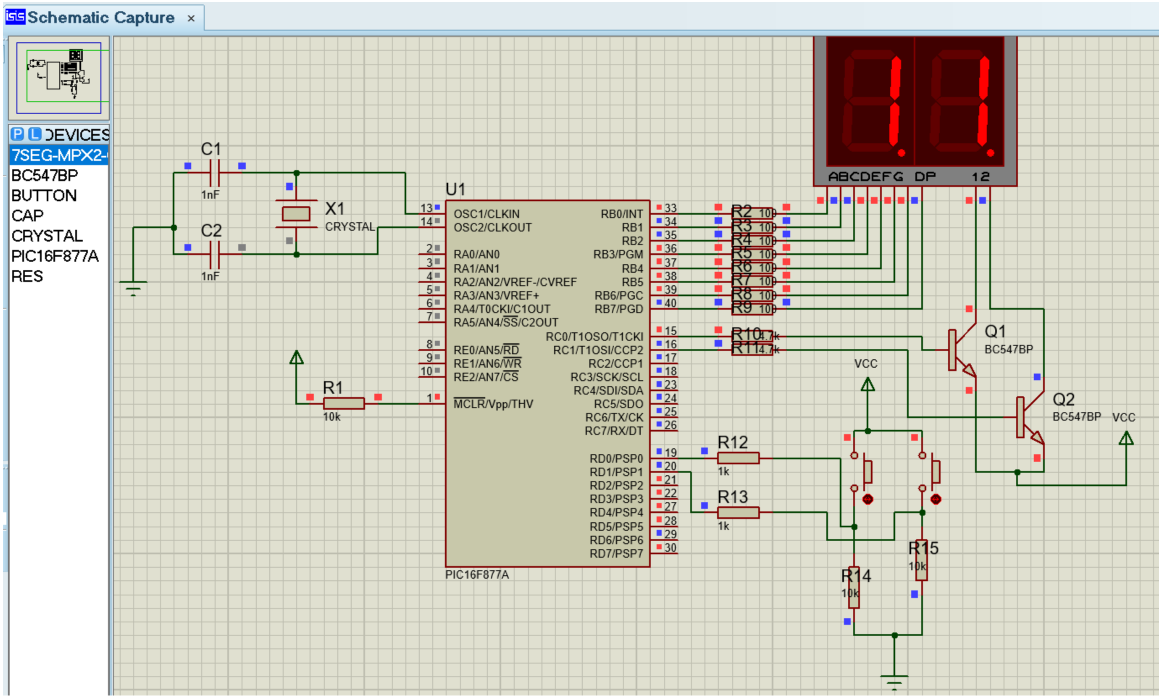
Circuit Diagram:



MikroC Code :

```
char array[]={0x3f, 0x06, 0x5b, 0x4f, 0x66, 0x6d, 0x7d, 0x07, 0x7f, 0x6f};
void main()
{
int i=0;
int j, Left_digit=0, Right_digit=0,count;
TRISB=0x00;
TRISC=0x00;
portb=0x00;
portc=0x00;
while(1)
{
Left_digit=i/10;
Right_digit=i%10;
for(j=0;j<50;j++)
{

portb=array[Left_digit];
portc.f0=0;
delay_ms(10);
portc.f0=1;
portc.f1=0;
portb=array[Right_digit];
delay_ms(10);
portc.f1=1;
}

if(i>99)
i=0;
else
i++;}}
```

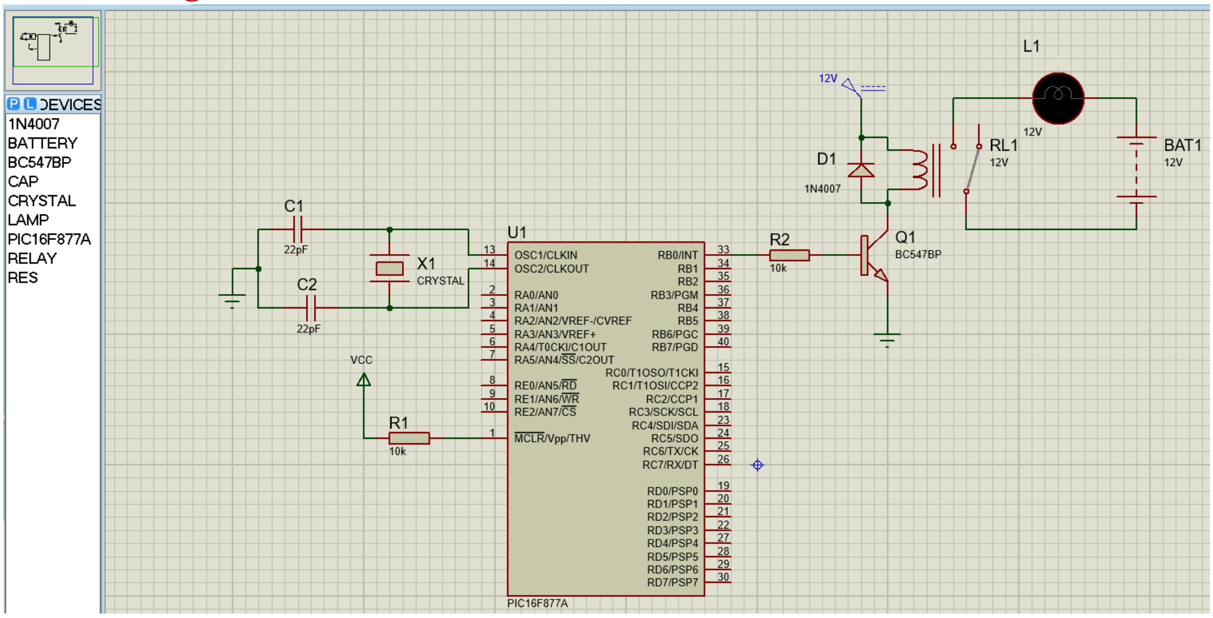
Circuit Diagram:



MikroC Code :

```
char arrayca[] = { 0x40, 0x79, 0x24, 0x30, 0x19, 0x12, 0x02, 0x78, 0x00, 0x10 }; // 0-9 segment codes
int tens,ones;
void main() {
   int number = 0;  // 0-99 number
// Set port direc ons
   TrisB = 0x00;   // PortB as output (data for 7-segment)
   TrisC = 0x00;   // PortC as output (digit select)
   TrisD = 0x03;   // PortD D0,D1 as input (bu ons)
 // Ini alize ports
   PortB = 0x00;
   PortC = 0x00;
while (1) {
      // ---- Bu on 1 pressed ? increment number ----
      if (PortD.F0 == 1) {
         Delay_ms(150);   // Debounce
         if (PortD.F0 == 1) {
            number++;
            if (number > 99) number = 0;  // Wrap 0-99
         }
      }
 // ---- Bu on 2 pressed ? decrement number ----
      if (PortD.F1 == 1) {
         Delay_ms(150);   // Debounce
         if (PortD.F1 == 1) {
            number--;
if (number < 0) number = 99; // Wrap 0-99
         }
      }
// ---- Separate digits ----
      tens = number / 10;
      ones = number % 10;
 // ---- Display tens digit ----
      PortC.F0 = 1;        // Enable first 7-segment
      PortB = arrayca[tens];
      Delay_ms(5);
      PortC.F0 = 0;        // Disable first 7-segment
// ---- Display ones digit ----
      PortC.F1 = 1;        // Enable second 7-segment
      PortB = arrayca[ones];
      Delay_ms(5);
      PortC.F1 = 0;        // Disable second 7-segment
   }
}
```

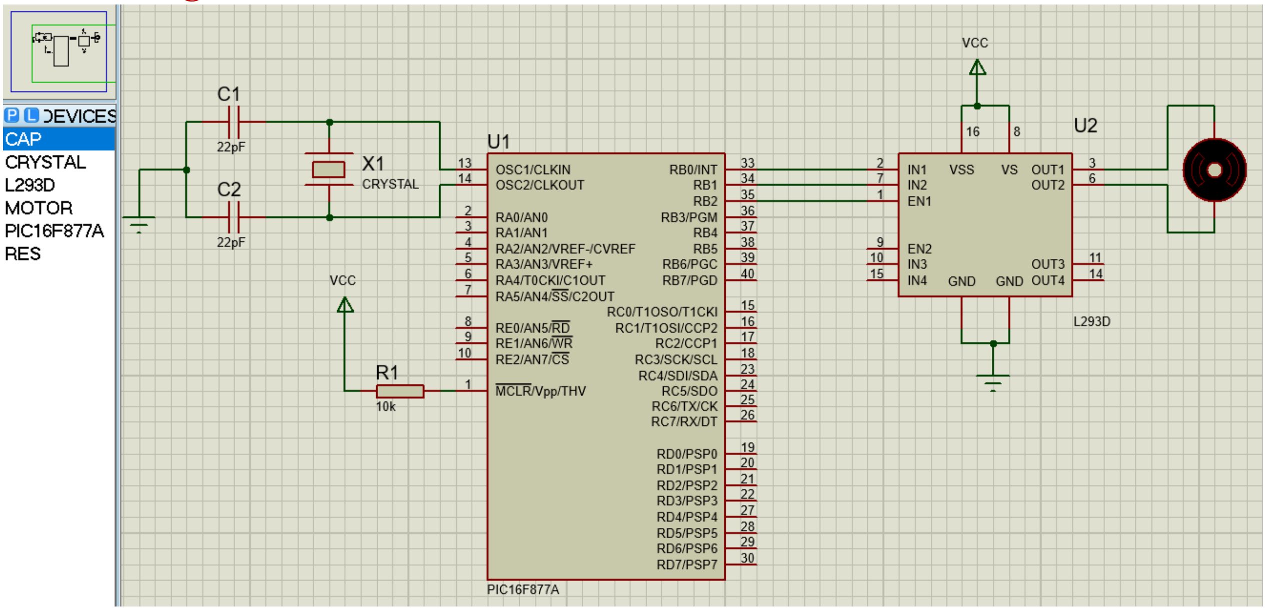
Circuit Diagram:



MikroC Code :

```
void main() {
        TRISB=0x00; //set port b as output
        portb=0x00; //initialize as 0
        while(1)
        {
              portb.f0=1;
              delay_ms(1000);
              portb.f0=0;
              delay_ms(1000);
     }
}
```

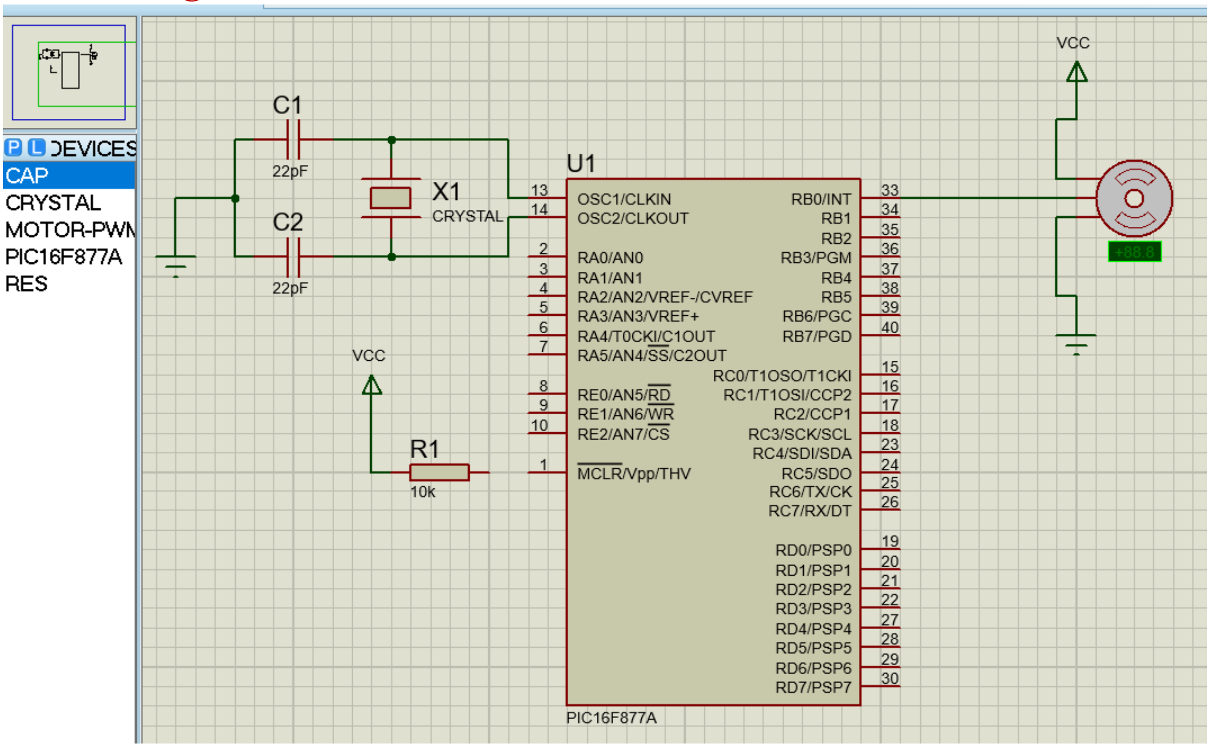
Circuit Diagram:



MikroC Code :

```
void main() {
 TRISB = 0x00;
 portb = 0x00;

 while(1){
 portb.f2=1;   //en1 pin active
 portb.f0 = 1;
 portb.f1 = 0;
 delay_ms(5000);

 portb.f0 = 0;
 portb.f1 = 1;
 delay_ms(5000);
 }
}
```

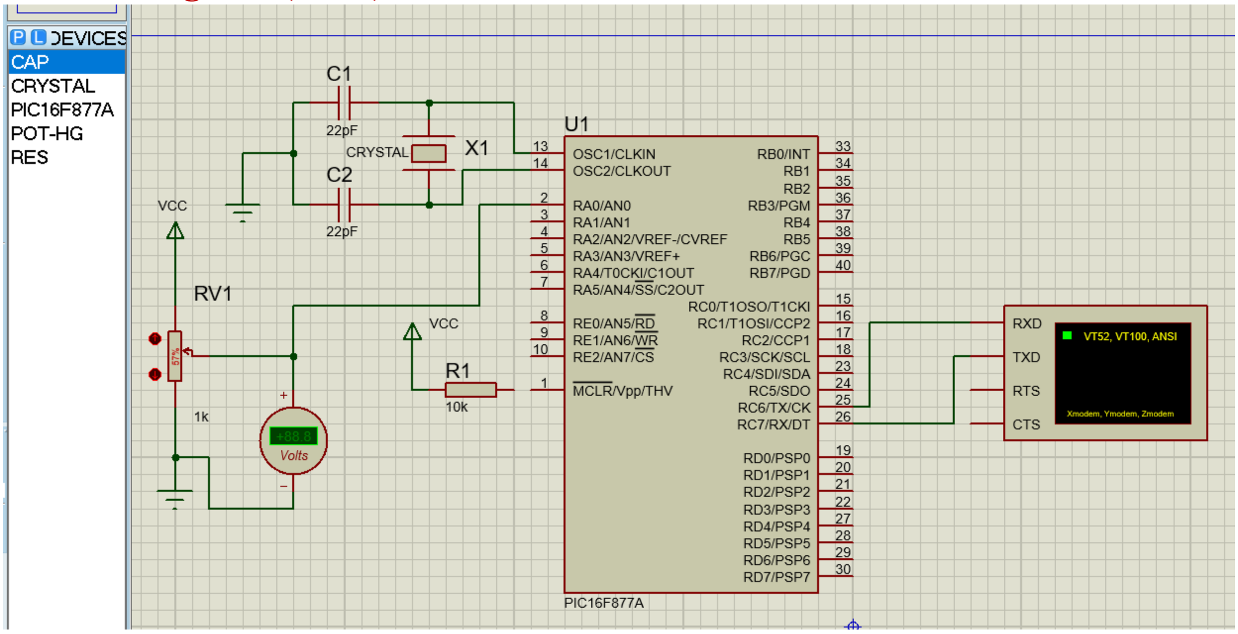
Circuit Diagram:



MikroC Code :

```
int i;
void main() {
   TRISB.F0 = 0;   // RB0 as output
   PORTB.F0 = 0;

   while(1){
      // LEFT 90 degree
      for(i=0; i<50; i++){
         PORTB.F0 = 1;
         Delay_us(1000);   // 1 ms pulse
         PORTB.F0 = 0;
         Delay_us(19000);  // remainder of 20 ms
      }
      Delay_ms(500);

      // STOP
      for(i=0; i<50; i++){
         PORTB.F0 = 1;
         Delay_us(1500);  // 1.5 ms pulse
         PORTB.F0 = 0;
         Delay_us(18500);
      }
      Delay_ms(500);

      // RIGHT 90 degree
      for(i=0; i<50; i++){
         PORTB.F0 = 1;
         Delay_us(2000);  // 2 ms pulse
         PORTB.F0 = 0;
         Delay_us(18000);
      }
      Delay_ms(500);

      // STOP
      for(i=0; i<50; i++){
         PORTB.F0 = 1;
         Delay_us(1500);  // 1.5 ms pulse
         PORTB.F0 = 0;
         Delay_us(18500);
      }
      Delay_ms(500);
   }
}
```

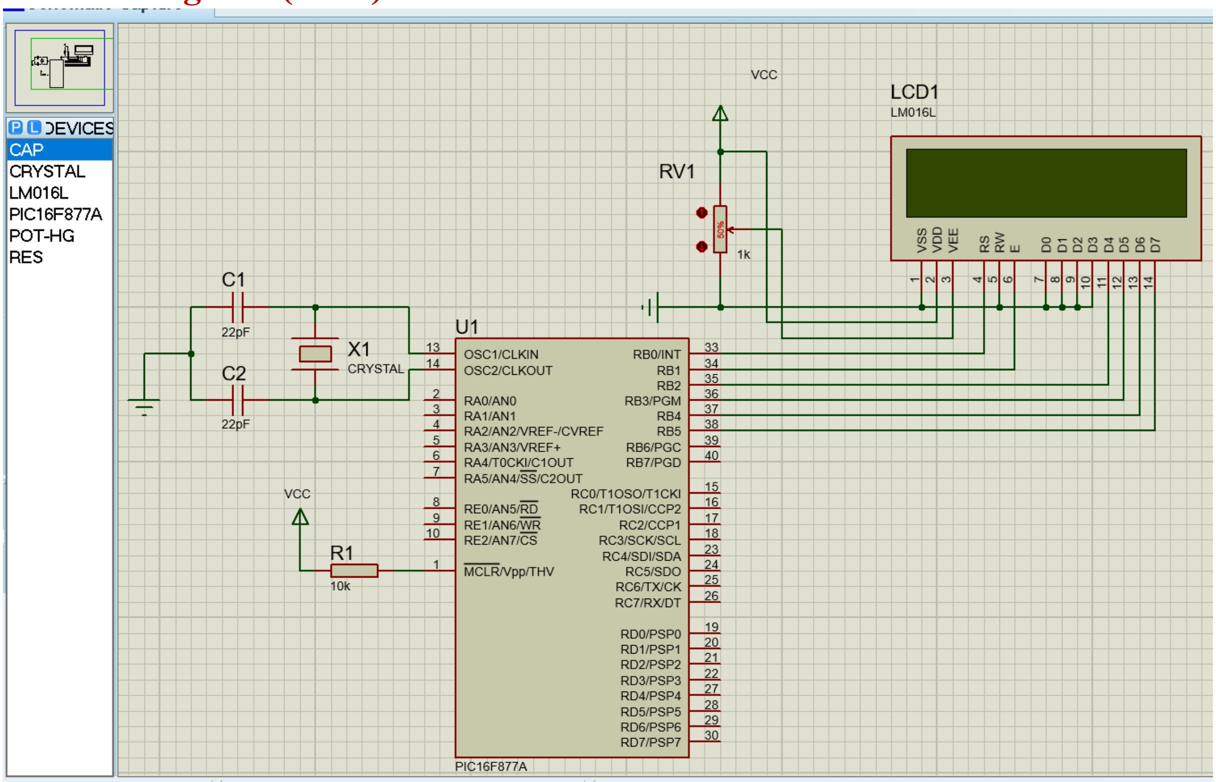
Circuit Diagram:(ADC)



MikroC Code :

```
unsigned int adc_value;
char txt[7];

void main() {

    ADCON1 = 0x80;        // AN0 analog
    TRISA = 0xFF;         // PORTA input (AN0 input)
    TRISC = 0x80;         // RC7 input, RC6 output

    UART1_Init(9600);     // UART 9600 baud
    Delay_ms(100);

    while(1)
    {
        adc_value = ADC_Read(0);

        IntToStr(adc_value, txt);  // Integer ? String

        UART1_Write_Text("ADC Value = ");
        UART1_Write_Text(txt);
        UART1_Write(13);   // Carriage return
        UART1_Write(10);   // New line

        Delay_ms(500);
    }
}
```

Circuit Diagram:(LCD)

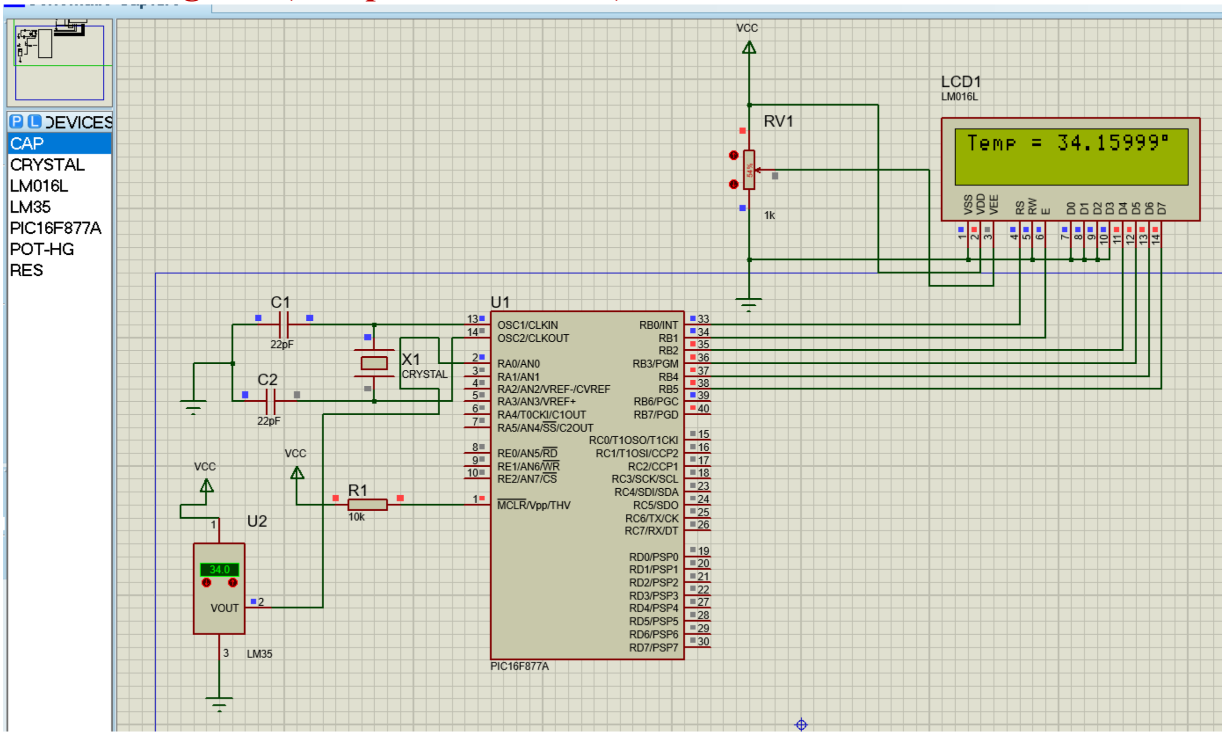


MikroC Code :

```
// LCD module connections
sbit LCD_RS at RB0_bit;
sbit LCD_EN at RB1_bit;
sbit LCD_D4 at RB2_bit;
sbit LCD_D5 at RB3_bit;
sbit LCD_D6 at RB4_bit;
sbit LCD_D7 at RB5_bit;

sbit LCD_RS_Direction at TRISB0_bit;
sbit LCD_EN_Direction at TRISB1_bit;
sbit LCD_D4_Direction at TRISB2_bit;
sbit LCD_D5_Direction at TRISB3_bit;
sbit LCD_D6_Direction at TRISB4_bit;
sbit LCD_D7_Direction at TRISB5_bit;
// End LCD module connections

char txt1[] = "Information & Communication Engineering";
char txt2[] = "ICE(PUST)";

char i;                    // Loop variable

void Move_Delay() {                // Function used for text moving
```

```
  Delay_ms(300);                      // we can change the moving speed here
}

void main(){
  ADCON1  = 0x06;                     // Configure AN pins as digital I/O
  CMCON = 0x07;

  Lcd_Init();
  Lcd_Cmd(_LCD_CLEAR);
  Lcd_Cmd(_LCD_CURSOR_OFF);

  Lcd_Out(1,1,txt1);
  Lcd_Out(2,5,txt2);

  Delay_ms(1000);

  while(1) {                         // Endless loop
    for(i=0; i<24; i++) {
    Lcd_Cmd(_LCD_SHIFT_LEFT);
    Move_Delay();
}

    /* for(i=0; i<8; i++) {             // Move text to the right 7 times
      Lcd_Cmd(_LCD_SHIFT_RIGHT);
      Move_Delay();
    }   */
  }
}
```

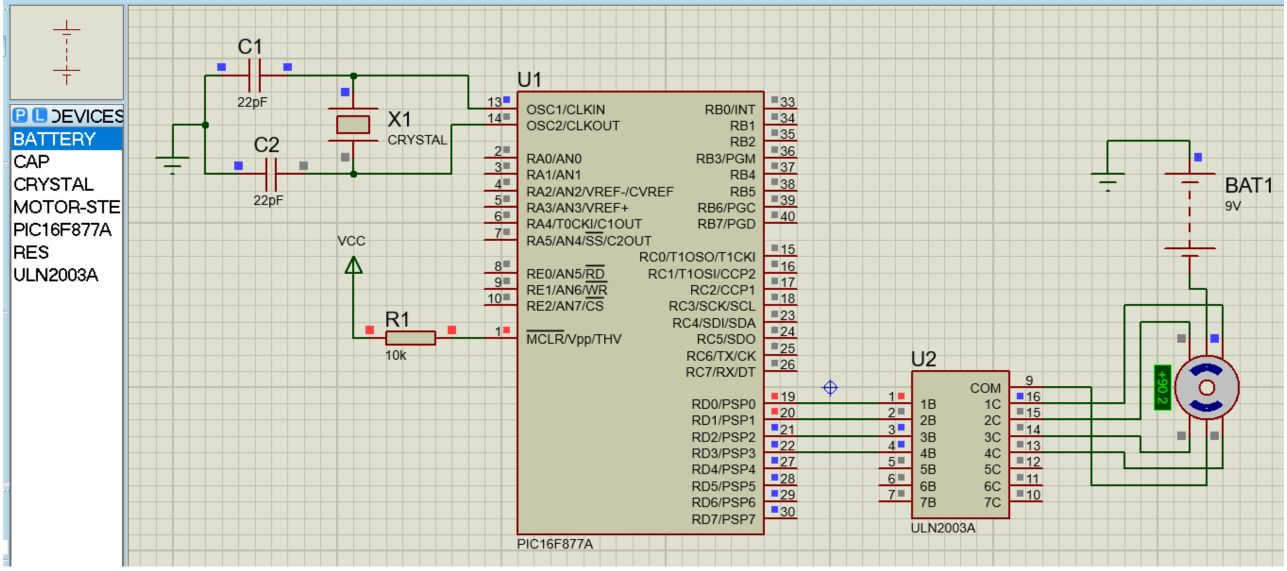
Circuit Diagram:(Temperature sensor)



MikroC Code :

```
// LCD module connections
sbit LCD_RS at RB0_bit;
sbit LCD_EN at RB1_bit;
sbit LCD_D4 at RB2_bit;
sbit LCD_D5 at RB3_bit;
sbit LCD_D6 at RB4_bit;
sbit LCD_D7 at RB5_bit;
sbit LCD_RS_Direction at TRISB0_bit;
sbit LCD_EN_Direction at TRISB1_bit;

sbit LCD_D4_Direction at TRISB2_bit;
sbit LCD_D5_Direction at TRISB3_bit;
sbit LCD_D6_Direction at TRISB4_bit;
sbit LCD_D7_Direction at TRISB5_bit;
// End LCD module connections
char display[16]="";
void main()
{
unsigned int result;
float volt,temp;
trisb=0x00;
trisa=0xff;
adcon1=0x80;
lcd_init();
lcd_cmd(_lcd_clear);
lcd_cmd(_LCD_CURSOR_OFF);
while(1)
{
result=adc_read(0);
volt=result*4.88;
temp=volt/10;
lcd_out(1,1,"Temp = ");
floattostr(temp,display);
lcd_out_cp(display);
lcd_chr(1,16,223); //print at pos(row=1,col=13) "°" =223 =0xdf
lcd_out_cp(" C"); //celcius
}
}
```

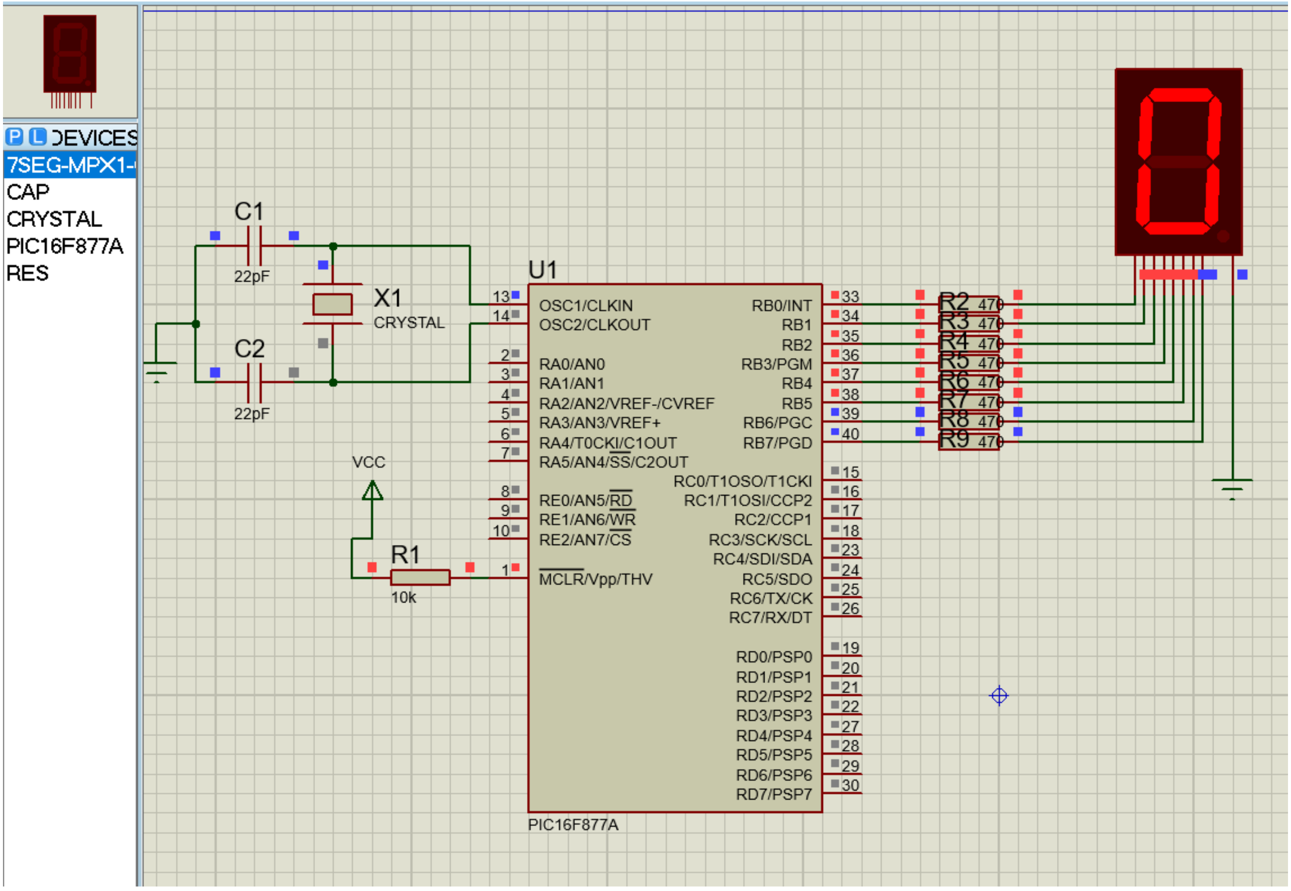
Circuit Diagram:(Stepper Motor)



MikroC Code :

```
void main()
{
 TRISD = 0x00;
 PORTD = 0x00;
 do
 {
  PORTD = 0b00000011;
  Delay_ms(500);
  PORTD = 0b00000110;
  Delay_ms(500);
  PORTD = 0b00001100;
  Delay_ms(500);
  PORTD = 0b00001001;
  Delay_ms(500);
 } while(1);
}
```

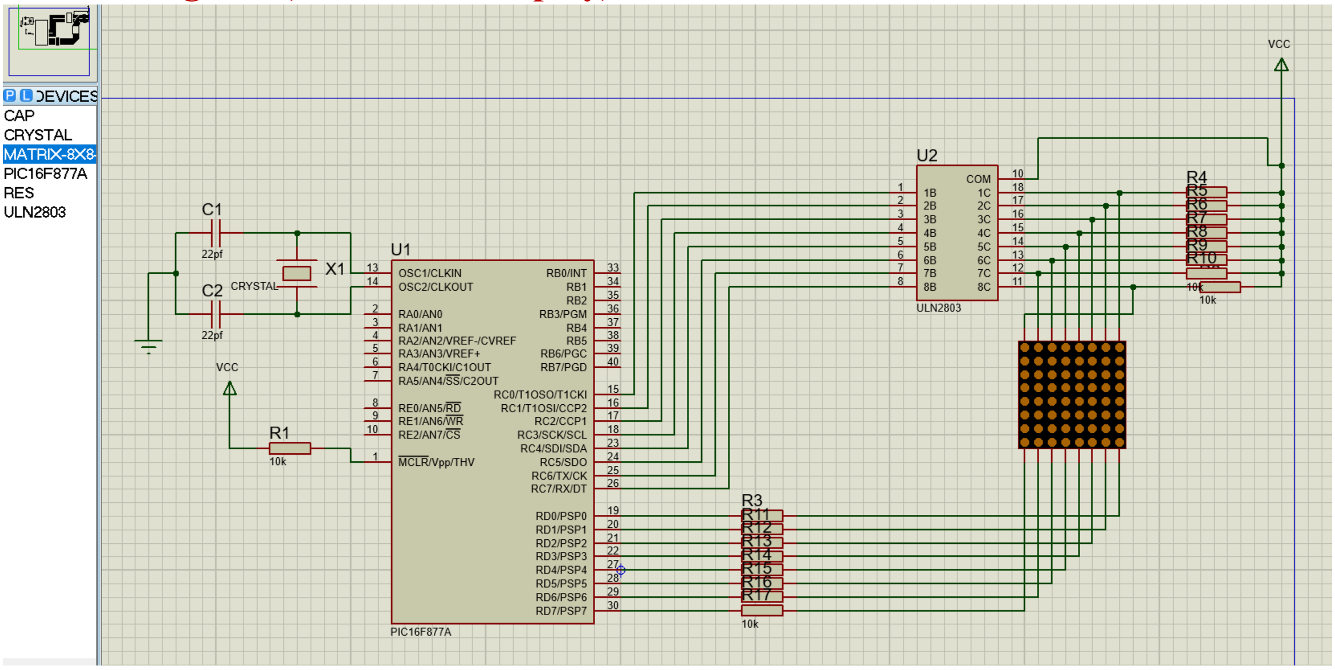
Circuit Diagram:(EEPROM)



MikroC Code :

```
unsigned char count;
unsigned char seg[] = {0x3F,0x06,0x5B,0x4F,0x66,0x6D,0x7D,0x07,0x7F,0x6F};
void main(){
  ADCON1 = 0x06;
  TRISB = 0x00;
  count = EEPROM_Read(0x00);
  if(count > 9) count = 0;
  while(1){
    PORTB = seg[count];
    Delay_ms(1000);
    count++;
    if(count > 9) count = 0;
    EEPROM_Write(0x00, count);
    Delay_ms(20);
  }
}
```

Circuit Diagram:(Dot Matrix Display)



MikroC Code :

```
void MSDelay(unsigned char Time)
{
unsigned char y,z;
for(y=0;y<Time;y++)
for(z=0;z<20;z++);
}
void main()
{
TRISC = 0x00;
TRISD = 0x00;
while(1)
{
PORTD = 0x80;
PORTC = 0x00;
MSDelay(10);
PORTD = 0x40;
PORTC = 0xff;
MSDelay(10);
PORTD = 0x20;
PORTC = 0xff;
MSDelay(10);
PORTD = 0x10;
PORTC = 0x18;
MSDelay(10);
PORTD = 0x08;
PORTC = 0x18;
MSDelay(10);
PORTD = 0x04;
PORTC = 0xff;
MSDelay(10);
PORTD = 0x02;
PORTC = 0xff;
MSDelay(10);
PORTD = 0x01;
PORTC = 0x00;
MSDelay(10);
}
}
```